

STATISTICS FORMULA SHEET – Bocconi University (Course 30001)

1. DESCRIPTIVE STATISTICS

Mean $\bar{x} = \sum x_i / n$; Weighted: $\sum x_k p_k$; Median $Me = (x_{(n/2)} + x_{(n/2+1)}) / 2$; Mode = most frequent.
Range = Max – Min; IQR = $Q3 - Q1$; Variance $s^2 = \sum (x_i - \bar{x})^2 / (n-1)$; SD $s = \sqrt{s^2}$; CV = $s / \bar{x}$.
Covariance $s_{xy} = \sum (x_i - \bar{x})(y_i - \bar{y}) / (n-1)$; Correlation $r = s_{xy} / (s_x s_y)$.

2. PROBABILITY

$P(A \cup B) = P(A) + P(B) - P(A \cap B)$; $P(A|B) = P(A \cap B) / P(B)$; Indep.: $P(A \cap B) = P(A)P(B)$.
 $E[X] = \sum x_i P(X = x_i)$; $Var(X) = E[X^2] - (E[X])^2$; $SD(X) = \sqrt{Var(X)}$.

3. RANDOM VARIABLES

Bernoulli(p): $P(X=x) = p^x (1-p)^{1-x}$, $E[X] = p$, $Var(X) = p(1-p)$.
Normal(μ, σ^2): $f(x) = 1/(\sqrt{2\pi\sigma^2}) e^{-\frac{(x-\mu)^2}{2\sigma^2}}$, $Z = (X-\mu)/\sigma$, $E[X] = \mu$, $Var(X) = \sigma^2$.
Linear transform: $Y = a + bX \rightarrow E[Y] = a + bE[X]$, $Var(Y) = b^2 Var(X)$.

4. SAMPLING DISTRIBUTIONS

$E[\bar{x}] = \mu$, $Var(\bar{x}) = \sigma^2/n$, $SE = \sigma/\sqrt{n}$; $E[\hat{p}] = p$, $Var(\hat{p}) = p(1-p)/n$, $SE = \sqrt{p(1-p)/n}$.

5. POINT ESTIMATION

Unbiased: $E[\hat{\theta}] = \theta$; Bias = $E[\hat{\theta}] - \theta$; Efficiency $\rightarrow$ smaller $Var(\hat{\theta})$.
Mean estimator $\bar{x} \rightarrow SE = \sigma/\sqrt{n}$ or $s/\sqrt{n}$.
Proportion estimator $\hat{p} = x/n \rightarrow SE = \sqrt{p(1-p)/n}$ or $\sqrt{\hat{p}(1-\hat{p})/n}$.
Difference of means (indep): $SE = \sqrt{s_x^2/n_x + s_y^2/n_y}$; pooled: $s_p^2 = [(n_x-1)s_x^2 + (n_y-1)s_y^2] / (n_x + n_y - 2)$; $SE = s_p \sqrt{1/n_x + 1/n_y}$.
Paired means: $D_i = X_i - Y_i$, $SE = s_D/\sqrt{n}$. Difference of proportions: $SE = \sqrt{\hat{p}(1-\hat{p})/n_1 + \hat{p}(1-\hat{p})/n_2}$.

6. NORMAL DISTRIBUTION FUNCTIONS (R)

$pnorm(q, mean, sd) = P(X \leq q)$; $1 - pnorm(q) = P(X > q)$; $qnorm(p, mean, sd) =$ percentile.

7. UBSTATS FUNCTIONS

`distr.table.x()` freq tables; `distr.plot.x()` graphs; `distr.summary.x(stats="central"/"dispersion")` $\rightarrow$ mode, median, mean, range, IQR, sd, var.
`distr.table.xy()`, `distr.plot.xy()` $\rightarrow$ two-variable tables/plots.

All symbols: μ (pop. mean), σ^2 (pop. var), s^2 (sample var), p (pop. proportion), $\hat{p}$ (sample prop.), SE (standard error).